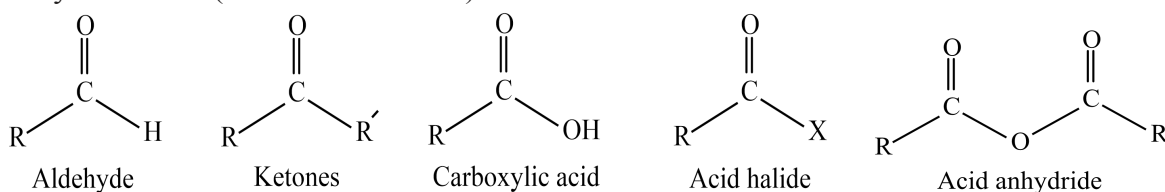


Unit 12:-Aldehydes, Ketones and Carboxylic Acids

The functional group containing carbon-oxygen double bond ($>C=O$) called carbonyl group and the compounds are called carbonyl compounds.

- (i) In aldehydes, the carbonyl group is bonded to a carbon and hydrogen.
- (ii) In the ketones, it is bonded to two carbon atoms.
- (iii) The carbonyl compounds in which carbonyl group is bonded to oxygen are known as carboxylic acids, and their derivatives (e.g. Esters, Anhydrides).
- (iv) The carbonyl compounds where $C=O$ group is attached to nitrogen are called amides.
- (v) The carbonyl compounds where $C=O$ group is attached to halogens are called acylhalides (or acid halides).



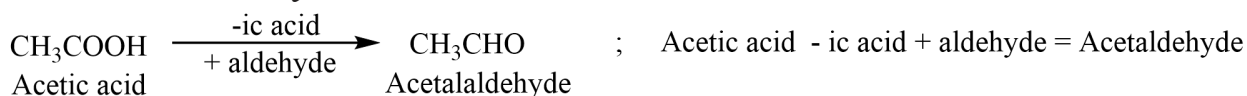
Nomenclature of aldehydes:

There are two systems of nomenclature of aldehydes and ketones.

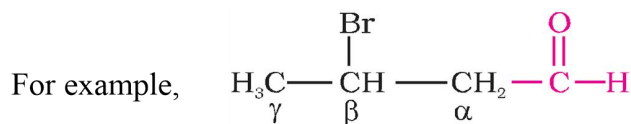
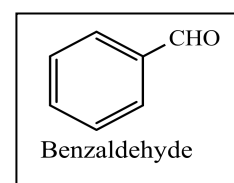
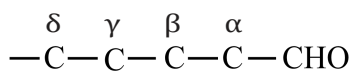
(a) **Common names:**

(b) **IUPAC names:**

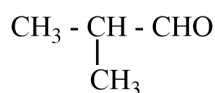
Common Names:- The common names of most aldehydes are derived from the common names of the corresponding carboxylic acids by replacing the ending *-ic* of acid with aldehyde. For example, CH_3CHO is derived from acetic acid (CH_3COOH) is named as **acetaldehyde**.



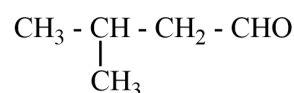
The location of the substituent (branch) in the carbon chain, if any, is indicated by Greek letters α , β , γ , δ , etc. The **α -carbon** being the one directly linked to the aldehyde ($-\text{CHO}$) group, β - carbon the next, and so on.



β -Bromobutyraldehyde



α -Methylpropionaldehyde



β -Methylbutyraldehyde

IUPAC Names:- The **IUPAC** names of open chain aliphatic **aldehydes** are derived from the names of the corresponding **alkanes** by replacing the ending *-e* with *-al*. i.e.,

Alkane - e + al = Alkanal e.g., Methane - e + al $\longrightarrow$ Methanal,

For example, HCH=O (Or HCHO)
Methanal

CH_3CHO
Ethanal

$\text{CH}_3\text{CH}_2\text{CHO}$
Propanal

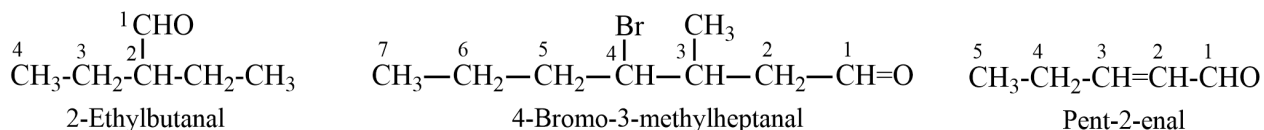
$\text{CH}_3 \text{---} \text{CH}_2 \text{---} \text{CH}_2 \text{---} \overset{\text{O}}{\parallel} \text{C---H}$
Butanal

 **Some Important Rules,**

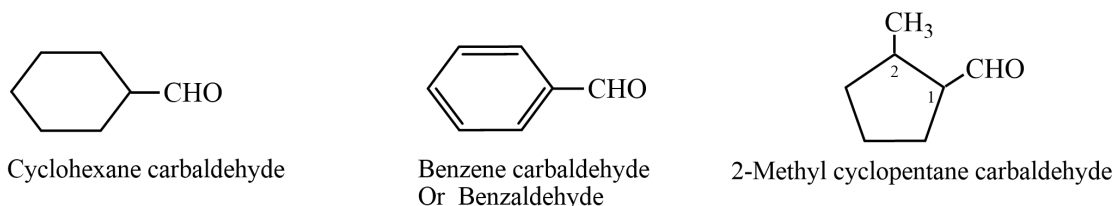
- (i) The longest carbon chain containing the $-\text{CHO}$ group is consider as parent chain.

- (ii) In case of substituted (or branched) aldehydes, the parent chain is numbered starting from the carbon of the aldehyde group (-CHO).
- (iii) The substituents are prefixed in alphabetical order along with numerals indicating their positions in the carbon chain.

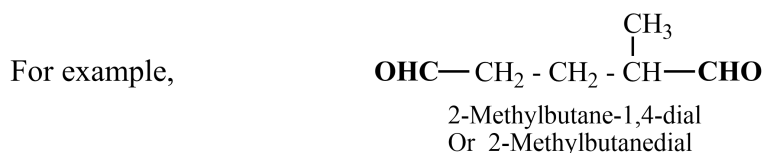
For example,



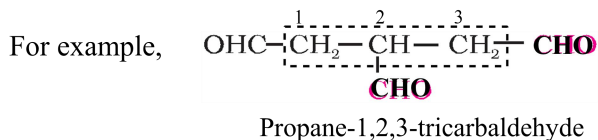
- (iv) When the aldehyde group is attached to a **ring**, the suffix **carbaldehyde** is added after the full name of the cycloalkane. The numbering of the ring carbon atoms start from the carbon atom attached to the aldehyde group. For example,



- (iv) If two -CHO groups are present in the compound the 'e' of parent **alkane** is not omitted and 'di' word is added before the suffix 'al'.

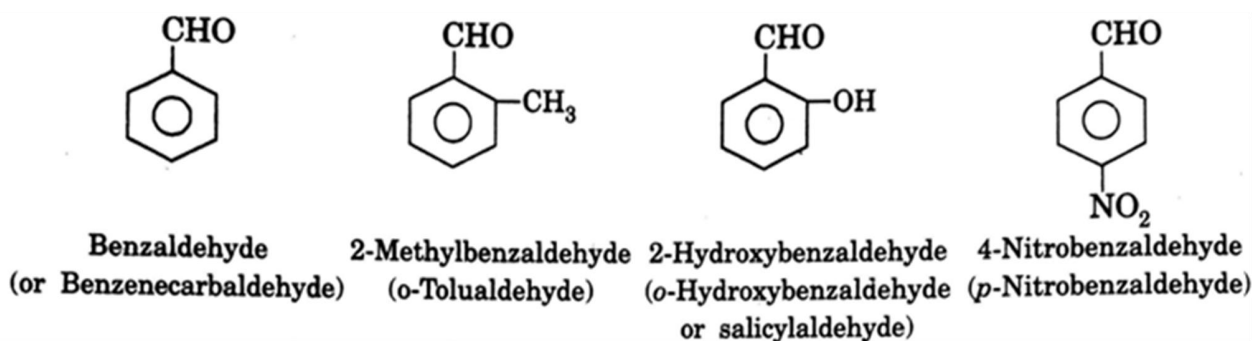


- ☞ If an unbranched chain is directly bonded to more than two -CHO groups then the 'C' atoms of the -CHO groups are not included in the parent chain and the suffix **carbaldehyde** is used instead of 'al'.

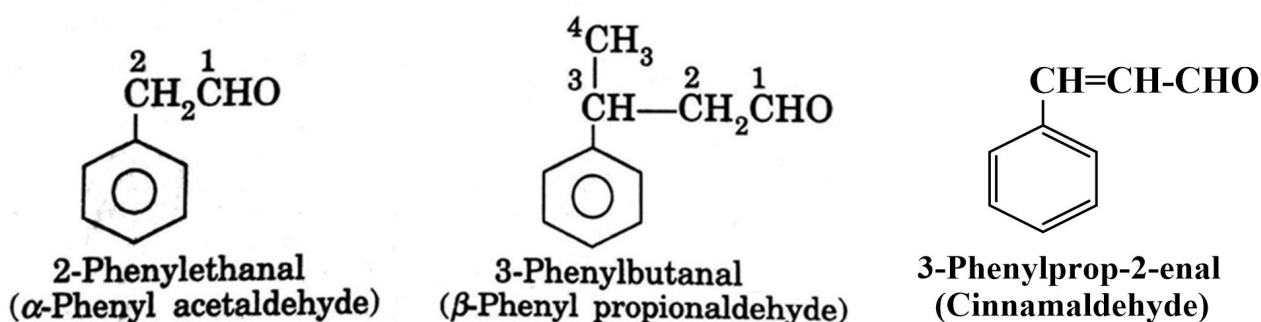


- ☞ In aromatic aldehydes, -CHO group is directly attached to the benzene ring and named as **benzene carbaldehyde**. However, the common name **benzaldehyde** is also accepted by IUPAC.

❖ IUPAC names of some important aromatic aldehydes:-



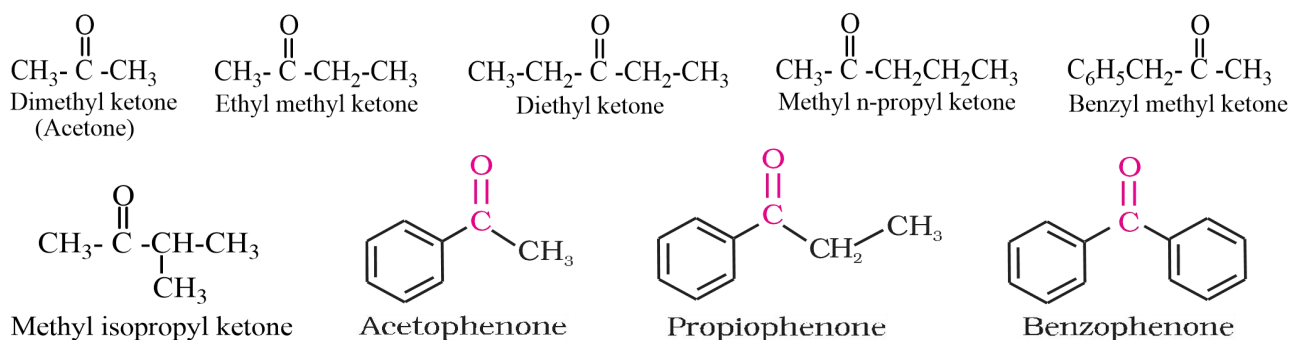
► The aldehydic group may also be present in the side chain. For example,



Nomenclature of Ketones:-

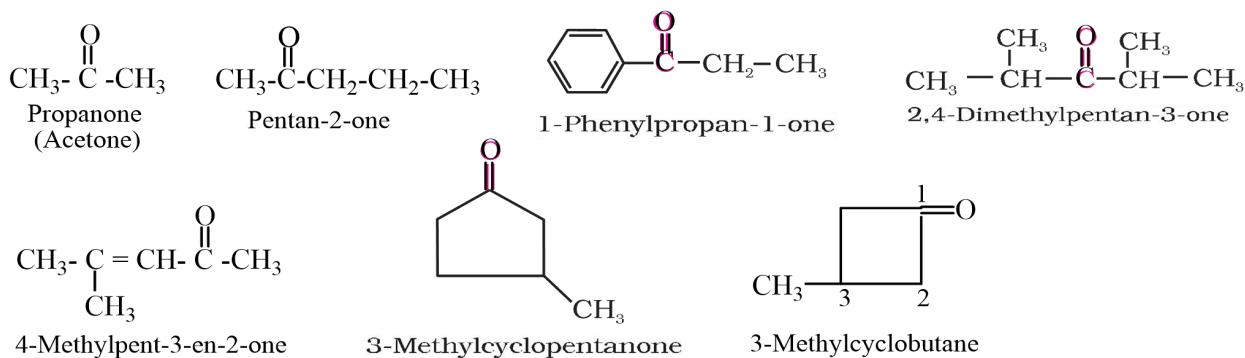
(a) **Common names**:- The common names of **ketones** are derived by naming two **alkyl** or **aryl** groups bonded to the carbonyl group ($>C=O$).

- ☞ The locations of substituents are indicated by Greek letters, α , α' , β , β' and so on beginning with the carbon atoms next to the carbonyl group, indicated as $\alpha\alpha'$.
- ☞ Some ketones have historical common names, the simplest **dimethyl ketone** is called **acetone**.
- ☞ **Alkyl phenyl ketones** are usually named by adding the acyl group as prefix to **phenone**. For example,

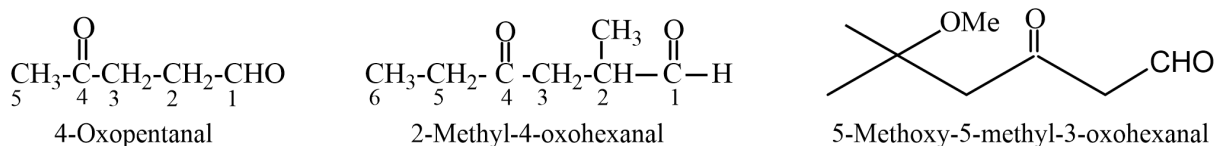


(b) **IUPAC names**:- The IUPAC names of open chain aliphatic **ketones** are derived from the names of the corresponding **alkanes** by replacing the ending 'e' with 'one'.

- ☞ The longest carbon chain is numbered starting from the carbon of the end nearer to the carbonyl group.
- ☞ The same applies to cyclic ketones, where the carbonyl carbon is numbered one.



Note:- If the compound contains both **aldehyde** and **ketone** groups, then **aldehyde** group is considered as **principal functional group** and the **ketone** group is regarded as **substituent** for which prefix 'oxo' is used along with a number to indicate its position. For example,



Intext Questions 12.1:- Write the structures of the following compounds.

- (i) α -Methoxypropionaldehyde (ii) 3-Hydroxybutanal
 (iii) 2-Hydroxycyclopentane carbaldehyde (iv) 4-Oxopentanal
 (v) Di-sec. butyl ketone (vi) 4-Fluoroacetophenone

Preparation of Aldehydes and Ketones:-

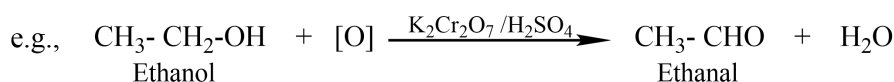
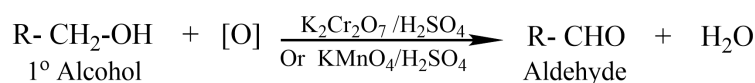
Some important methods for the preparation of aldehydes and ketones are as follows:

A. Preparation of both Aldehydes and Ketones:

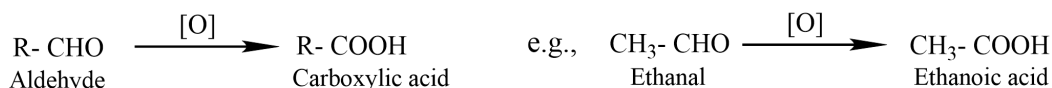
1. By oxidation of alcohols:

Aldehydes and ketones are generally prepared by oxidation of primary and secondary alcohols respectively with normal oxidizing agents such as $\text{K}_2\text{Cr}_2\text{O}_7$, KMnO_4 , CrO_3 (chromium oxide) and sulphuric acid. e.g.,

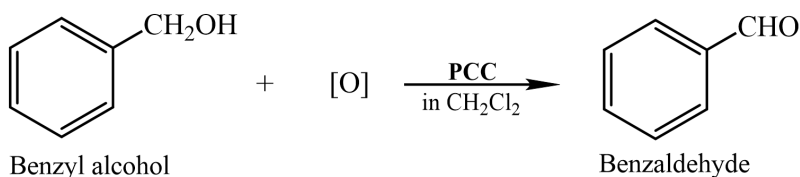
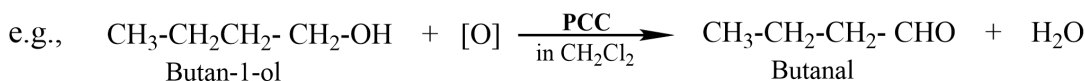
➤ 1° Alcohols give aldehydes,



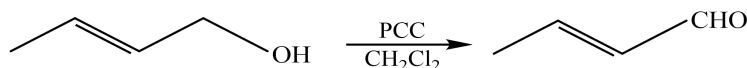
☞ Aldehydes undergo readily oxidation to carboxylic acids.



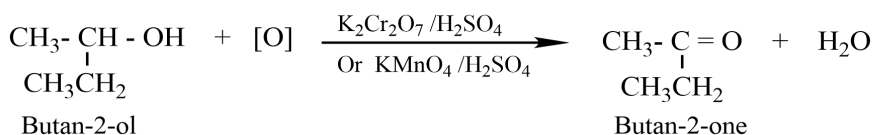
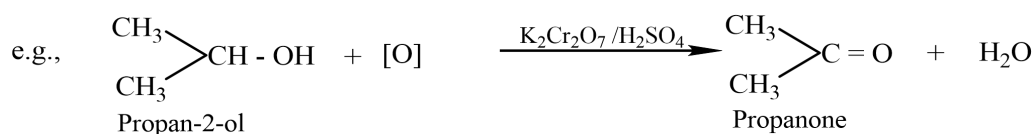
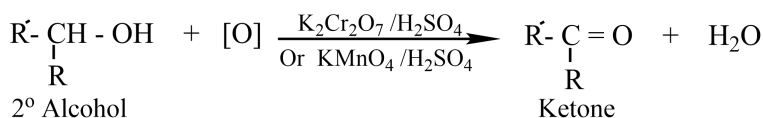
✓ Therefore, 1° alcohols can be converted to aldehyde stage by treating with mild oxidizing agents like **Pyridinium chlorochromate** ($\text{C}_5\text{H}_5\text{NH}^+\text{CrO}_3\text{Cl}^-$), which is abbreviated as **PCC**.

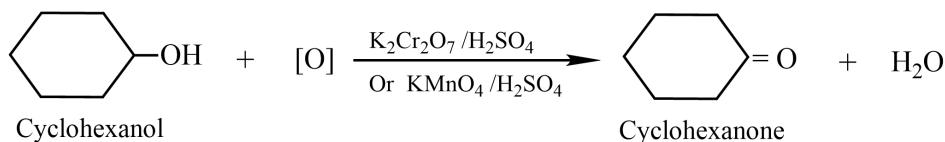


☞ It may be noted that PCC also does not attack carbon-carbon double bond,

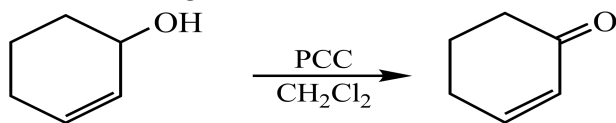


➤ 2° Alcohols give ketones,





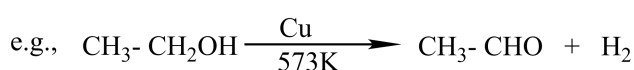
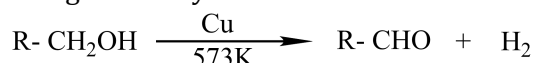
✓ PCC can also be used to get ketones from 2° alcohols,



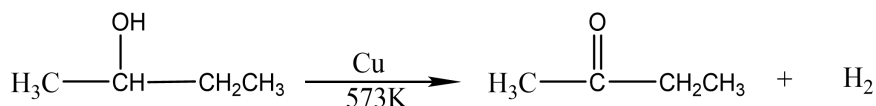
2. By catalytic dehydrogenation of alcohols:

This method is suitable for volatile alcohols and is of industrial application. In this method alcohol vapours are passed over heavy metal catalysts (Ag or Cu) at 573K. Primary and secondary alcohols give aldehydes and ketones, respectively.

☞ 1°-alcohols give aldehydes:

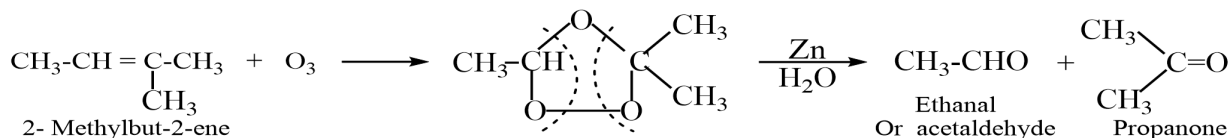
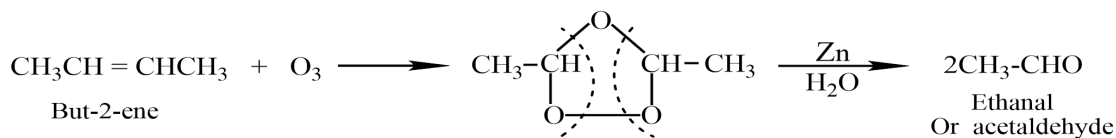
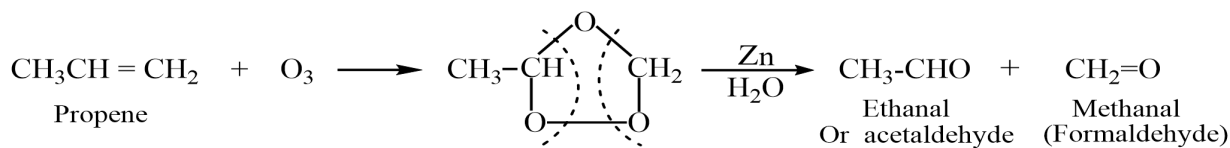
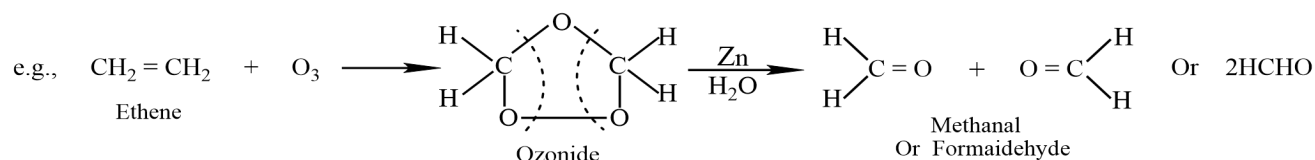
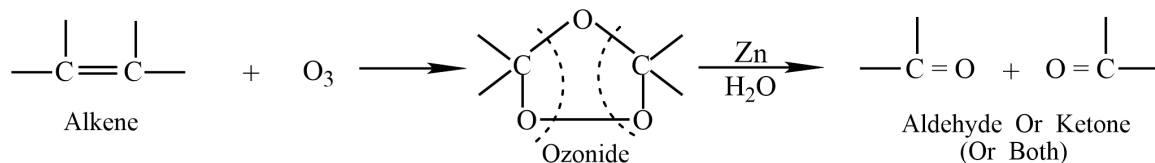


☞ 2°-alcohols give ketones:

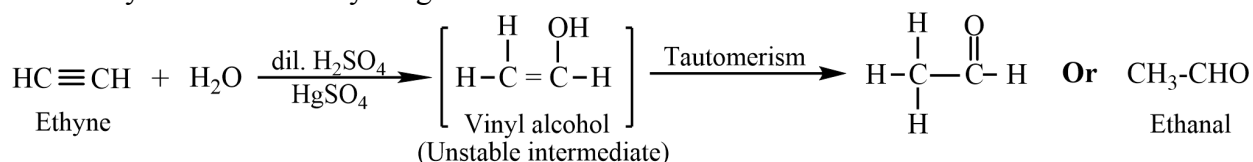


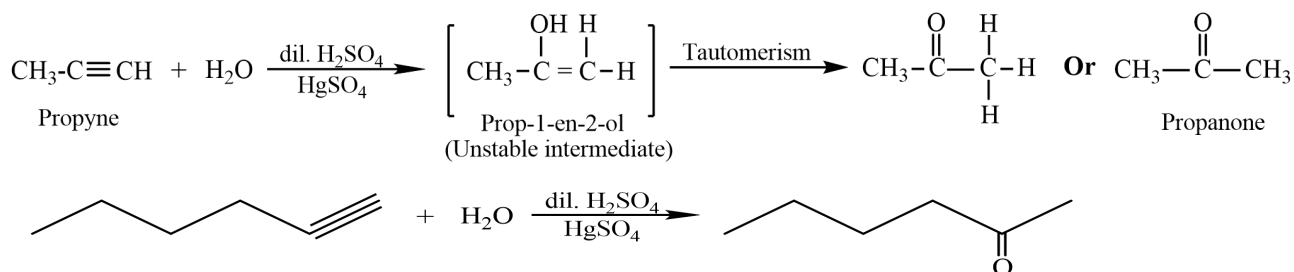
3. From hydrocarbons:

(i) **By ozonolysis of alkenes:-** Ozonolysis of alkenes followed by reaction with zinc dust and water gives aldehydes, ketones or a mixture of both depending on the substitution pattern of the alkene.

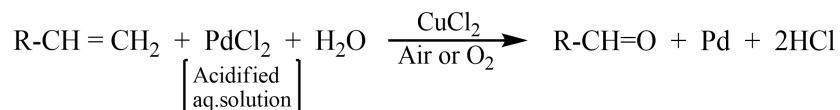


(ii) **By hydration of alkynes:** Addition of water to ethyne in the presence of dil. H₂SO₄ and HgSO₄ gives acetaldehyde. All other alkynes give ketones in this reaction.

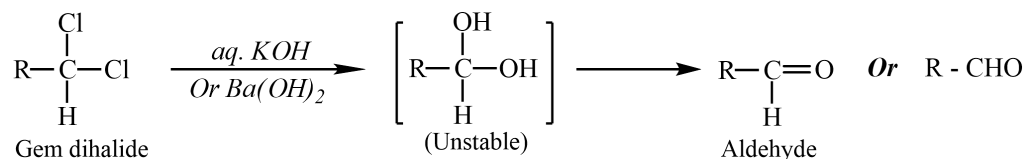




(iii) By Wacker's Process:

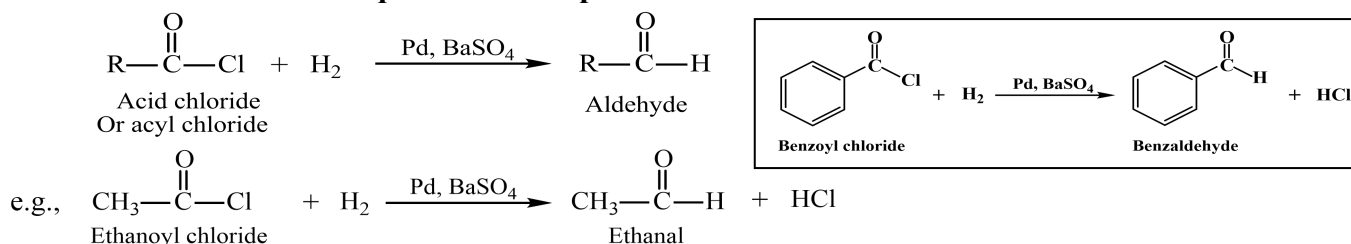


(iv) From 'Gem' Dihalides:

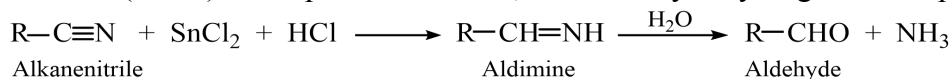


B. Preparation of Aldehydes only:

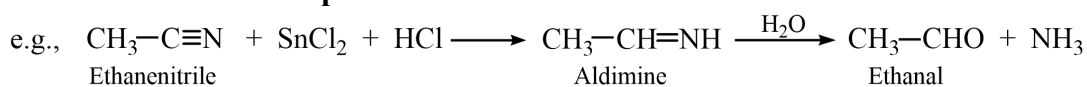
1. From acyl chloride (acid chloride): Acid chlorides can be converted into aldehydes by catalytic hydrogenation in the presence of 'Pd' catalyst supported over BaSO₄ and poisoned by the addition of small amount of quinoline or sulphur. This reaction is called **Rosenmund reduction**.



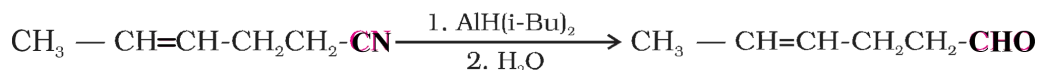
2. From Nitriles (Alkyl Cyanides) and Esters: Nitriles are reduced to corresponding imine with stannous chloride (SnCl₂) in the presence of HCl, which on hydrolysis give corresponding aldehyde.



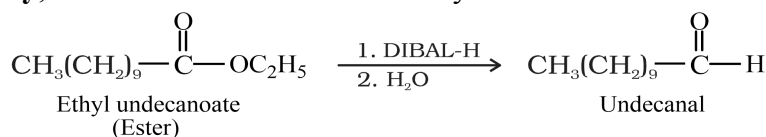
➤ This reaction is called **Stephen reaction**.



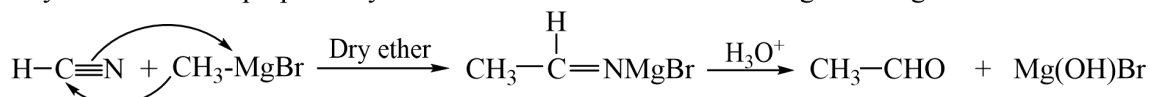
☞ **Alternatively**, nitriles are selectively reduced by diisobutylaluminium hydride, [DIBAL-H or AlH(i-Bu)₂] to imines followed by hydrolysis to aldehydes.



☞ **Similarly**, esters are also reduced to aldehydes with DIBAL-H.



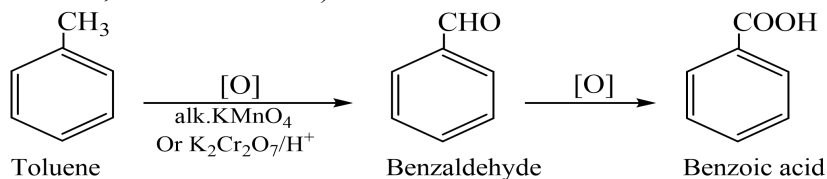
● Aldehydes can also be prepared by the reaction between HCN and Grignard reagents:



3. From Hydrocarbons (Preparation of Aromatic Aldehydes):

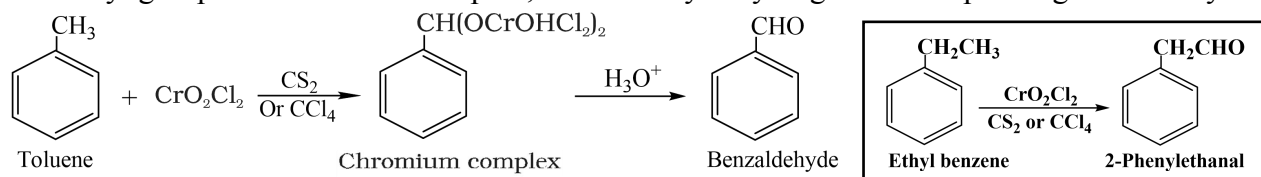
Aromatic aldehydes (i.e., benzaldehyde and its derivatives) are prepared from aromatic hydrocarbons by the following methods:

(i) By oxidation of methylbenzene: Strong oxidising agents (e.g., acidified or alkaline KMnO_4 , acidified $\text{K}_2\text{Cr}_2\text{O}_7$, conc. HNO_3 etc) oxidise toluene and its derivatives to benzoic acids.



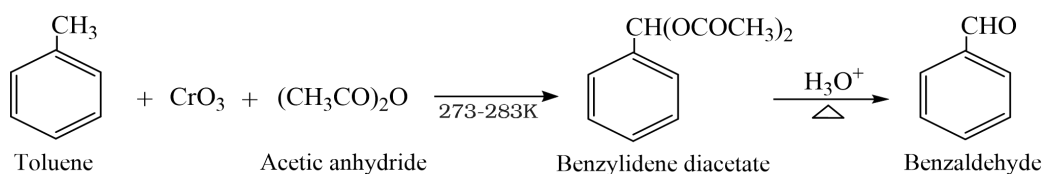
☛ **However**, it is possible to stop the oxidation at the aldehyde stage with suitable reagents that convert the methyl group to an intermediate that is difficult to oxidize further. The following methods are used for this purpose.

(a) Oxidation of methylbenzene by Chromyl chloride (CrO_2Cl_2): Chromyl chloride oxidises methyl group to a chromium complex, which on hydrolysis gives corresponding benzaldehyde.

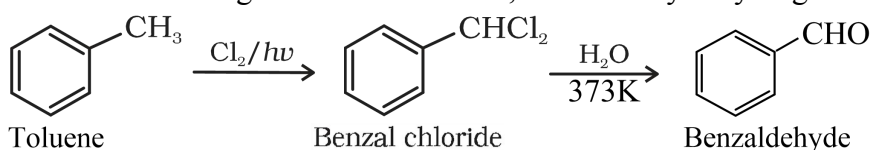


➤ This reaction is called **Etard reaction**.

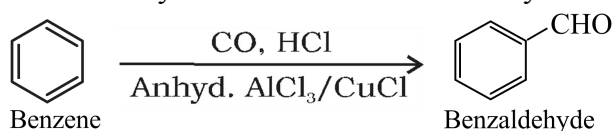
(b) Oxidation of methylbenzene by chromic oxide (CrO_3) in acetic anhydride: Toluene or substituted toluene is converted to benzylidene diacetate on treating with chromic oxide in acetic anhydride. The benzylidene diacetate can be hydrolysed to corresponding benzaldehyde with aqueous acid.



(ii) By side chain chlorination of methylbenzene followed by hydrolysis: Side chain chlorination of toluene gives benzal chloride, which on hydrolysis gives benzaldehyde.

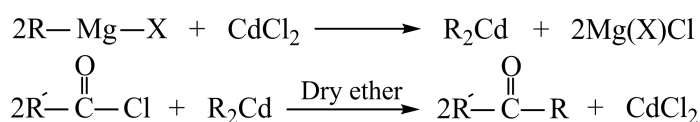


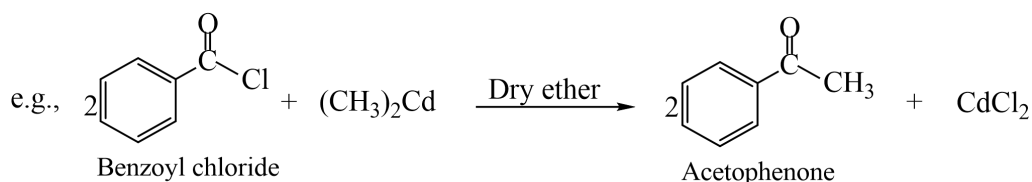
(iii) By Gatterman – Koch reaction: When benzene or its derivative is treated with carbon monoxide and hydrogen chloride in the presence of anhydrous aluminium chloride or cuprous chloride, it gives benzaldehyde or substituted benzaldehyde.



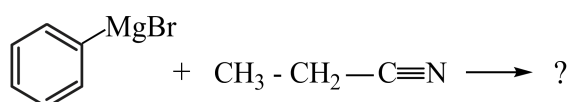
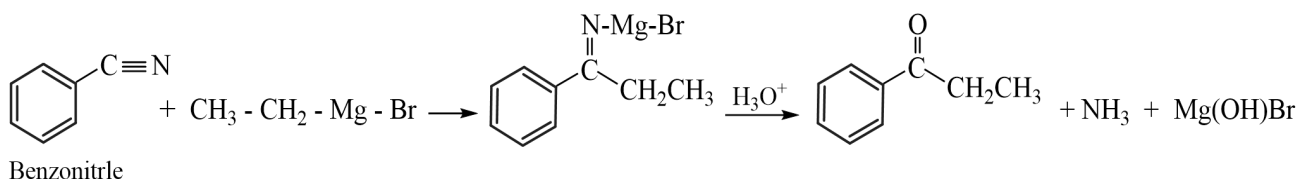
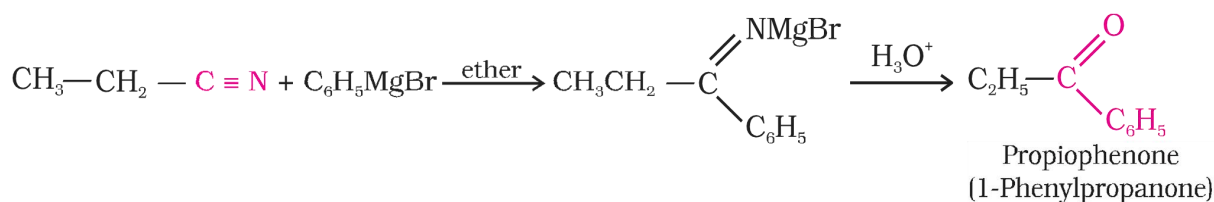
C. Preparation of Ketones only:

1. From acyl chlorides: Treatment of acyl chlorides with dialkylcadmium, prepared by the reaction of cadmium chloride with Grignard reagent, gives ketones.

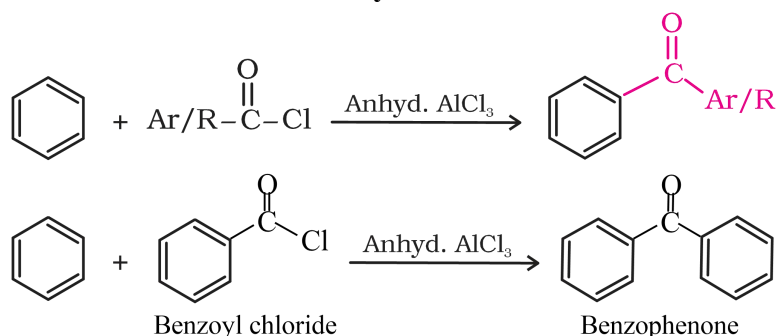




2. From nitriles: Treatment of a nitrile with Grignard reagent followed by hydrolysis yields a ketone.



3. From benzene or substituted benzenes: When benzene or substituted benzene is treated with acid chloride in the presence of anhydrous aluminium chloride, it affords the corresponding ketone. This reaction is known as **Friedel-Crafts acylation** reaction.



Example 12.1: Give names of the reagents to bring about the following transformations:

- | | |
|---|------------------------------------|
| (i) Hexan-1-ol to hexanal | (ii) Cyclohexanol to cyclohexanone |
| (iii) <i>p</i> -Fluorotoluene to <i>p</i> -fluorobenzaldehyde | (iv) Ethanenitrile to ethanol |
| (v) Allyl alcohol to propenal | (vi) But-2-ene to ethanol |

Solution:-

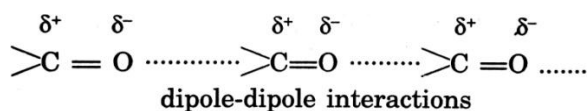
Physical Properties:

- ☞ Methanal is a gas at room temperature.
- ☞ Ethanal is a volatile liquid.
- ☞ Other aldehydes and ketones are liquid or solid at room temperature.

1. Boiling Points:

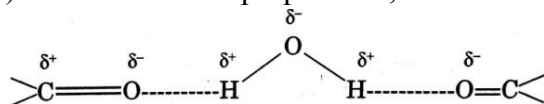
Question: Why are the boiling points of aldehydes and ketones are higher than hydrocarbons and ethers of comparable molecular masses?

Answer: It is due to weak molecular association in aldehydes and ketones arising out of the dipole-dipole interactions.



Note: The boiling points of aldehydes and ketones are lower than those of alcohols of similar molecular masses due to absence of intermolecular hydrogen bonding.

2. Solubility: The lower members of aldehydes and ketones such as methanal, ethanal and propanone are miscible (soluble) with water in all proportions, because they form hydrogen bond with water.



However, the solubility of aldehydes and ketones decreases rapidly on increasing the length of alkyl chain. All aldehydes and ketones are fairly soluble in organic solvents like benzene, ether, methanol, chloroform, etc.

Note: The lower aldehydes have sharp pungent odours. As the size of the molecule increases, the odour becomes less pungent and more fragrant. In fact, many naturally occurring **aldehydes** and **ketones** are used in the **blending of perfumes** and **flavouring agents**.

Question: Arrange the following compounds in the increasing order of their boiling points:



Answer: The increasing order of boiling points of the given compounds is as follows:



Intext Question: Arrange the following compounds in increasing order of their boiling points.



CHEMICAL PROPERTIES: Aldehydes and ketones are highly reactive and having same functional group, they undergo similar chemical reactions. The reactions of aldehydes and ketones can be discussed in the following categories:

A. Nucleophilic addition reactions.

B. Nucleophilic addition-elimination reactions.

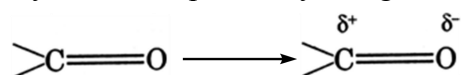
C. Reduction reactions.

D. Oxidation reactions.

E. Miscellaneous reactions.

A. Nucleophilic addition reactions: Aldehydes and ketones undergo nucleophilic addition reactions due to presence of highly reactive polar carbonyl ($>\text{C}=\text{O}$) group due to higher electronegativity of 'O' atom than 'C' atom. As a result, the electrons between carbon and oxygen are more towards oxygen atom.

Consequently, the carbonyl carbon is positively charged while the oxygen is negatively charged.



(i) Mechanism of nucleophilic addition reactions:

A nucleophile ($:\text{Nu}^-$) attacks the electrophilic 'C' atom ('+'ly charged) of the polar carbonyl group ($>\text{C}=\text{O}$) from a direction approximately perpendicular to the plane of sp^2 hybridised orbitals of

carbonyl carbon. The hybridization of carbon changes from sp^2 to sp^3 in this process, and a tetrahedral alkoxide intermediate is produced.

This intermediate, then captures a proton (H^+) from the reaction medium to give the electrically neutral product. The net result is addition of Nu^- and H^+ across the carbon oxygen double bond as shown below.

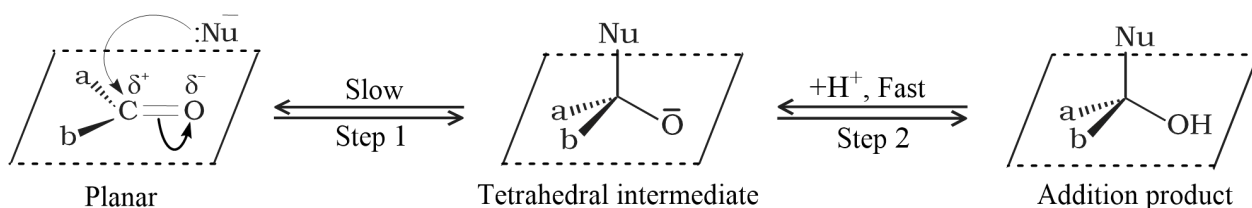


Fig.: Nucleophilic attack on carbonyl carbon

(ii)

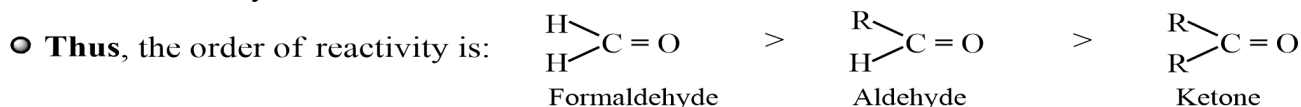
Reactivity:

Question:- Why are the Aldehydes generally more reactive than ketones in nucleophilic addition reactions?

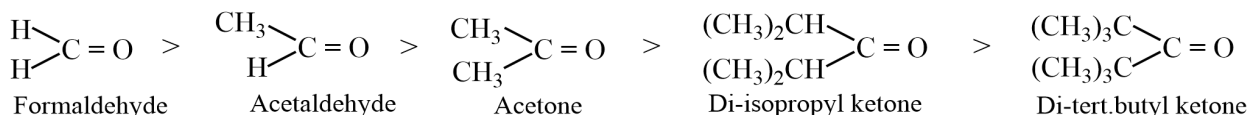
Answer: Aldehydes are generally more reactive than ketones in nucleophilic addition reactions due to **steric** and **electronic reasons**.

☞ **Sterically**, the presence of two relatively large substituents (alkyl groups) in ketones hinders the approach of nucleophile to carbonyl carbon than in aldehydes having only one such substituent.

☞ **Electronically**, aldehydes are more reactive than ketones because two alkyl groups reduce the **electrophilicity** (i.e., electron deficiency of 'C' atom) of the carbonyl group more effectively by **+I effect** than in aldehydes.

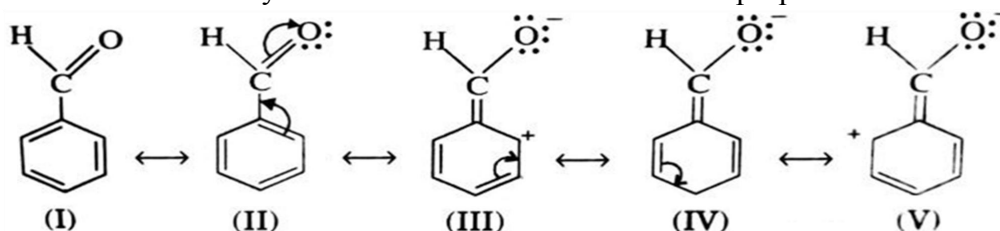


For example, the reactivity order of some aldehydes and ketones are;



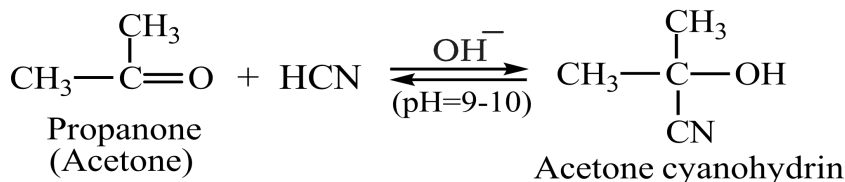
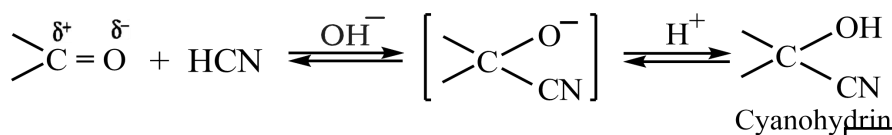
Question: Would you expect benzaldehyde to be more reactive or less reactive in nucleophilic addition reactions than propanal? Explain your answer. [NCERT example 12.3]

Solution: The carbon atom of the carbonyl group of benzaldehyde is less electrophilic than carbon atom of the carbonyl group present in propanal. It is because of the **electron releasing (+I effect)** and **resonance effect** of benzene ring as shown below. As a result, the polarity of the carbonyl group is reduced in benzaldehyde and hence it is less reactive than propanal.

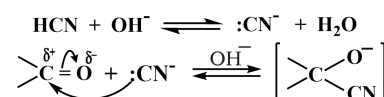


(iii) Some important examples of nucleophilic addition reactions:

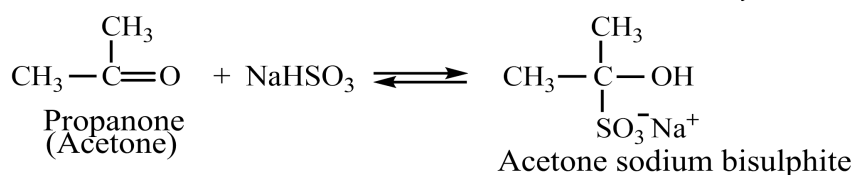
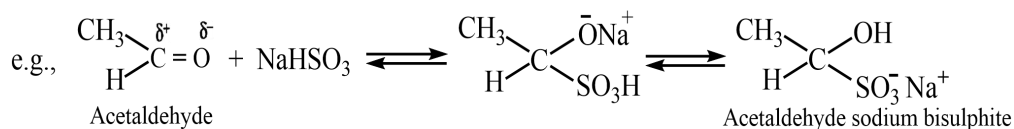
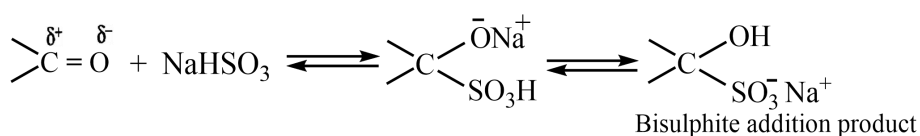
(a) Addition of hydrogen cyanide (HCN): Aldehydes and ketones react with hydrogen cyanide (HCN) to yield cyanohydrins.



☞ This reaction occurs very slowly with pure HCN. Therefore, it is catalysed by a base and the generated cyanide ion (CN^-) being a stronger nucleophile readily adds to carbonyl compounds to yield corresponding cyanohydrins.

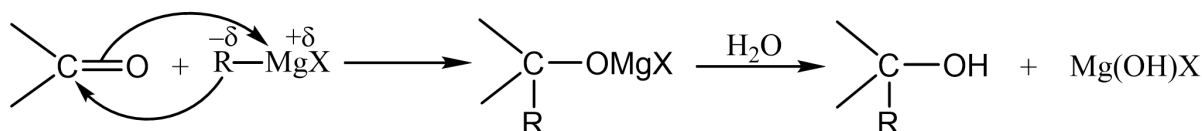


(b) Addition of sodium hydrogensulphite: Sodium hydrogensulphite adds to aldehydes and ketones to form the addition products.

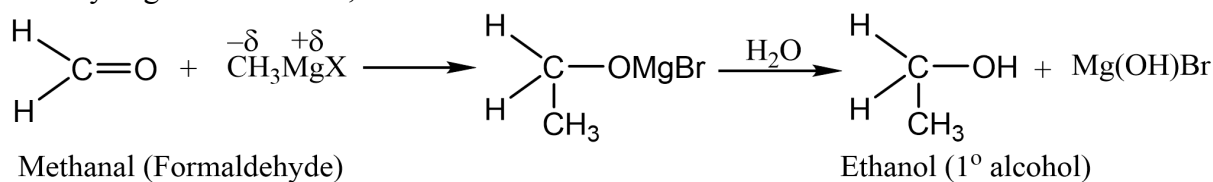


☞ The hydrogensulphite addition compound is water soluble and can be converted back to the original carbonyl compound by treating it with dilute mineral acid or alkali. Therefore, these are useful for separation and purification of aldehydes.

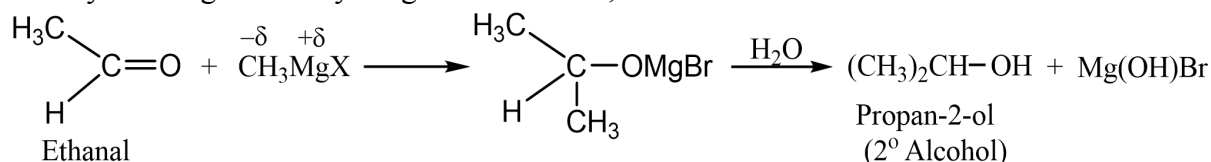
(c) Addition of Grignard reagents (R-MgX): The addition of Grignard reagent to the carbonyl group form an adduct, hydrolysis of which yields an alcohol.



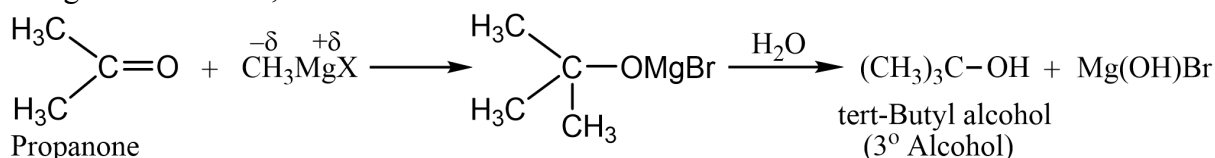
☞ Formaldehyde gives 1^o alcohol,



☞ Acetaldehyde or higher aldehydes give 2^o-alcohols,

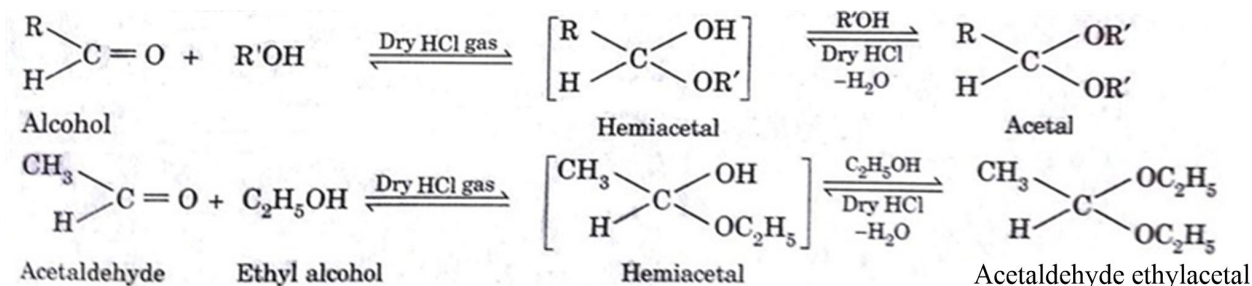


☞ Ketones give 3^o-alcohols,

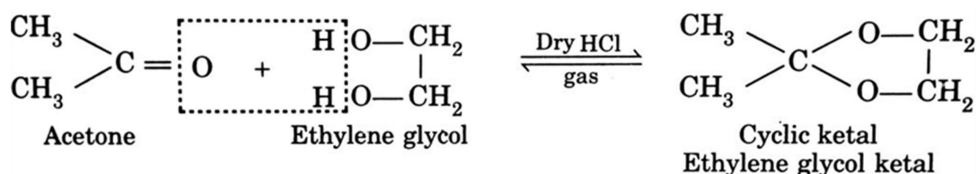


(d) Addition of alcohols: Aldehydes react with one equivalent of monohydric alcohol in the presence of dry hydrogen chloride to yield alkoxyalcohol intermediate, known as **hemiacetals**, which

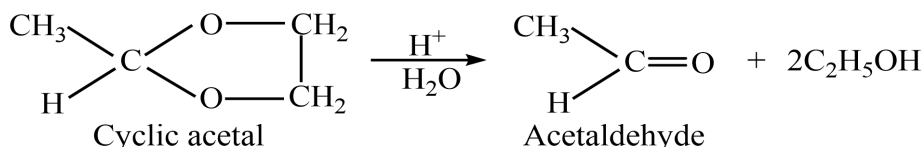
further react with one more molecule of alcohol to give a **gem-dialkoxy** compound known as **acetal** as shown in the reaction.



☞ **Ketones** do not react with monohydric alcohol but react with **ethylene glycol** (dihydric alcohol) under similar conditions to form cyclic products known as **ethylene glycol ketals**.



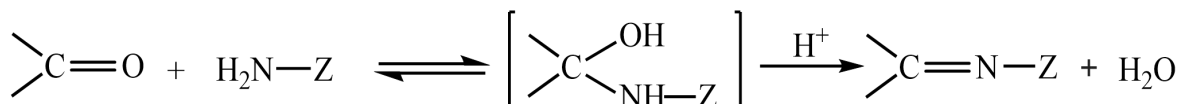
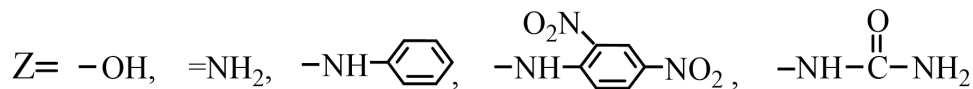
Note: Acetals and ketals are hydrolysed with aqueous mineral acids to yield corresponding aldehydes and ketones respectively. For example,



B. Nucleophilic addition-elimination reactions; Addition of ammonia and its derivatives:

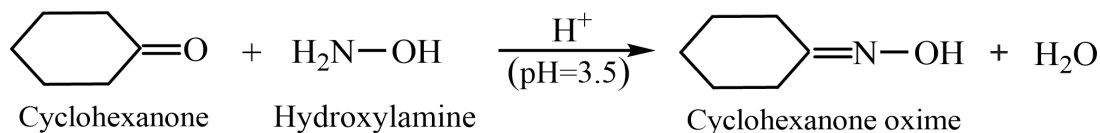
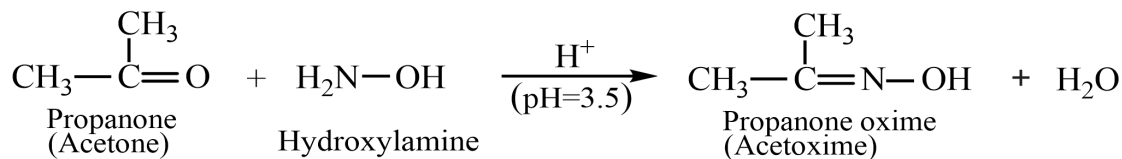
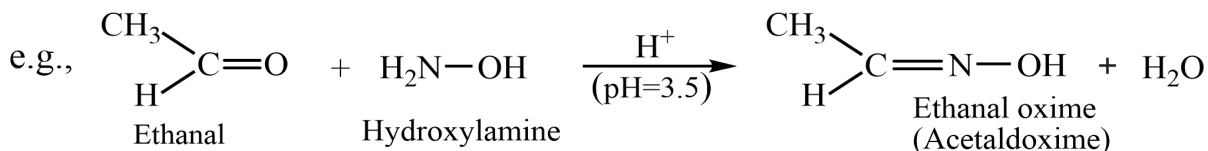
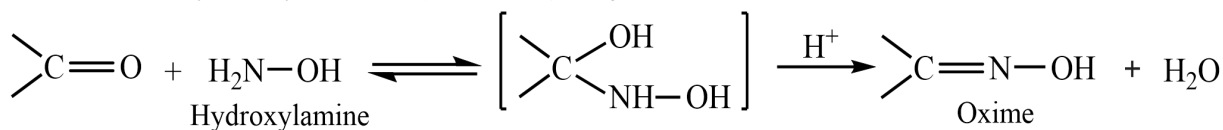
Aldehydes and ketones react with ammonia and a number of its derivatives ($\text{H}_2\text{N-Z}$) in *weakly acidic medium* to form compounds containing C=N double bonds with the elimination of H_2O molecule.

Where,

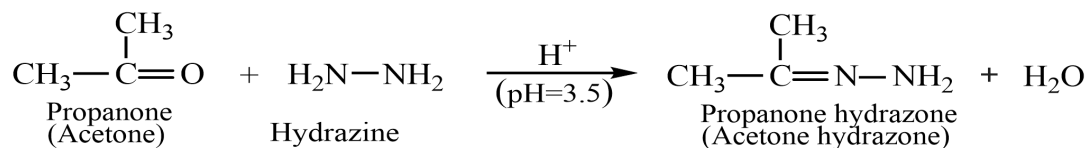
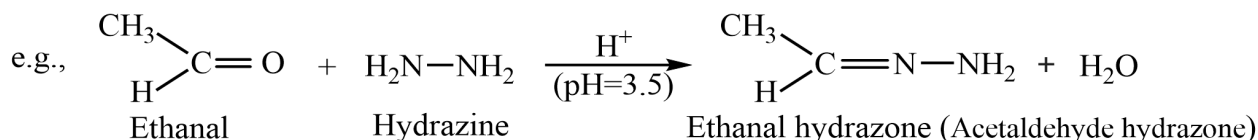
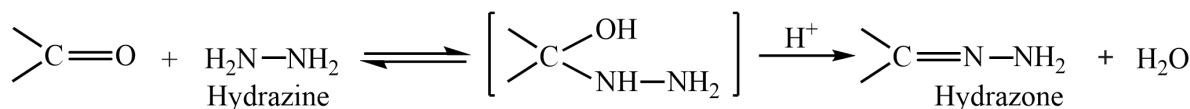


1. Reaction with some ammonia derivatives:

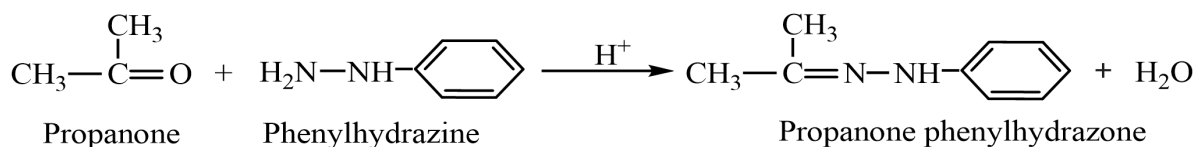
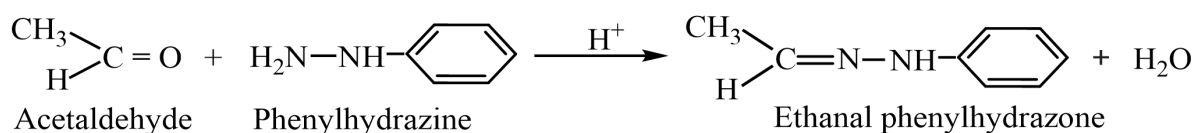
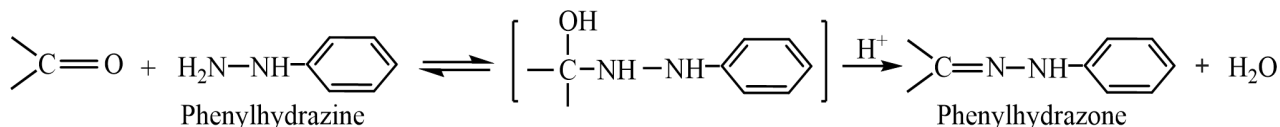
(i) Reaction with hydroxylamine (NH_2OH) to form oximes:



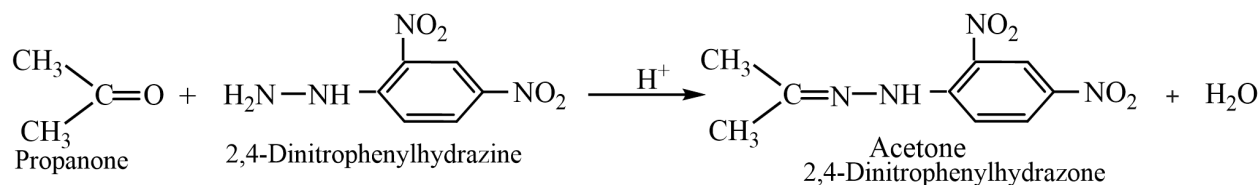
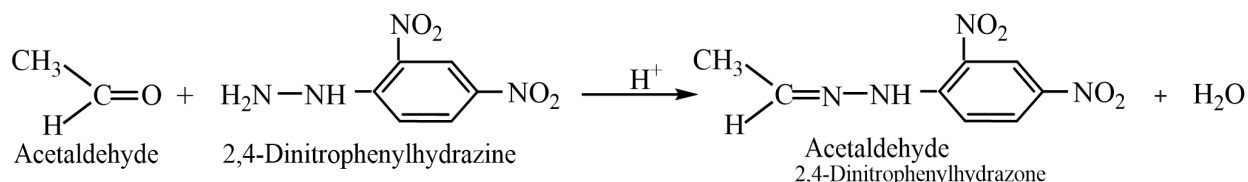
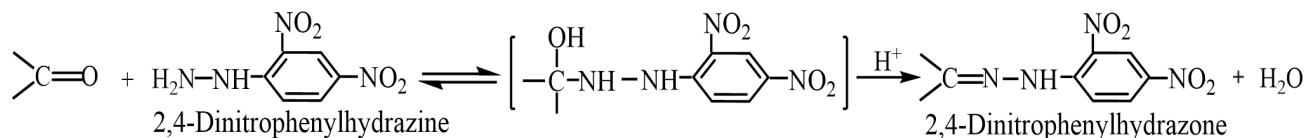
(ii) Reaction with hydrazine ($\text{H}_2\text{N-NH}_2$) to form hydrazone:



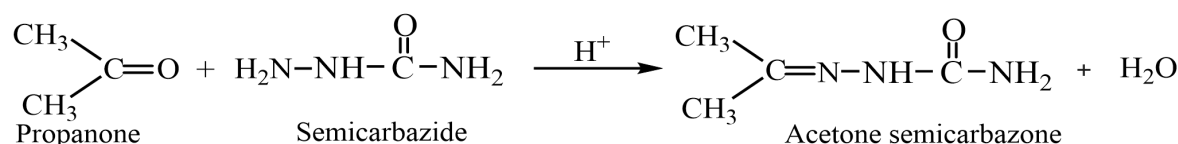
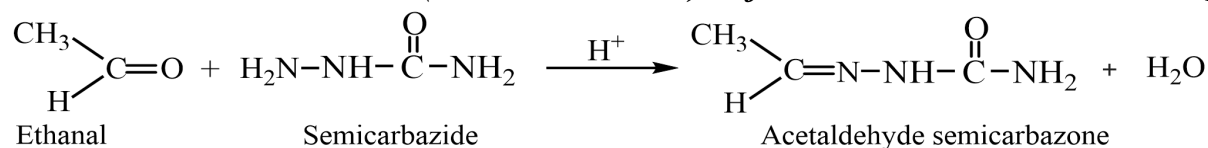
(iii) Reaction with phenylhydrazine ($\text{H}_2\text{N}-\text{NH}-\text{C}_6\text{H}_5$) to form phenylhydrazone:

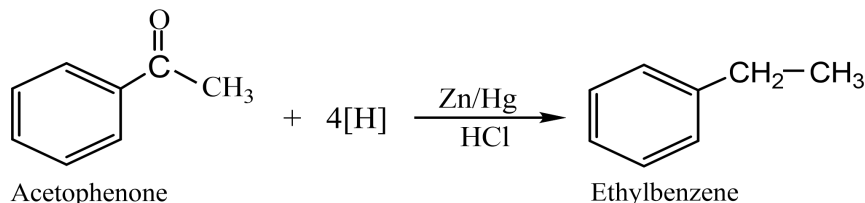
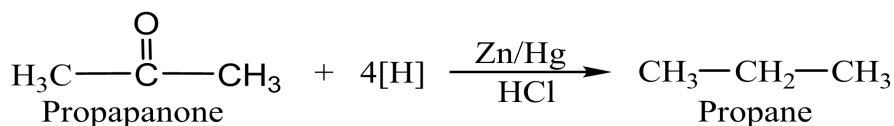
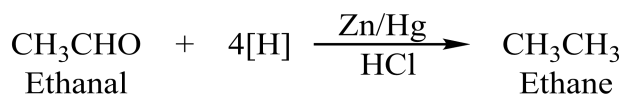


(iv) Reaction with 2,4-Dinitrophenylhydrazine (DNP) to form 2,4-dinitrophenylhydrazone:

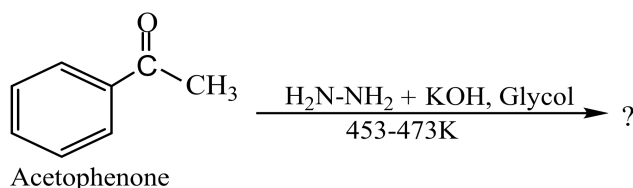
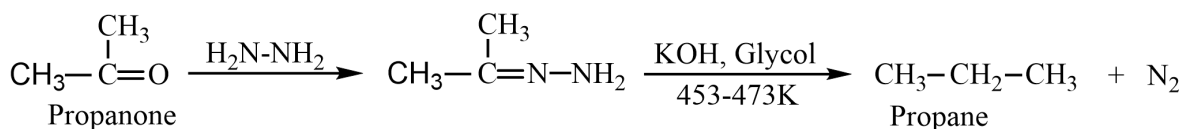
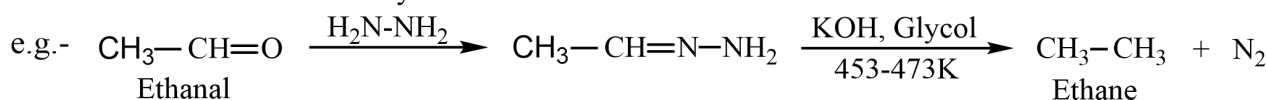
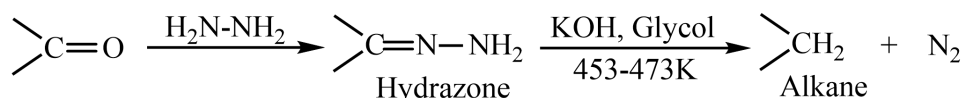


(v) Reaction with semicarbazide ($\text{H}_2\text{N}-\text{NHCONH}_2$) to form semicarbazone: For example,

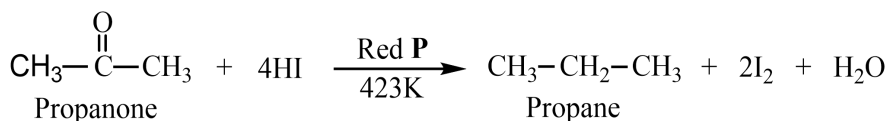




(b) Wolff-Kishner reduction: Aldehydes and ketones are treated with *hydrazine* ($\text{H}_2\text{N}-\text{NH}_2$) followed by heating with NaOH or KOH in *ethylene glycol* at 453-473K.



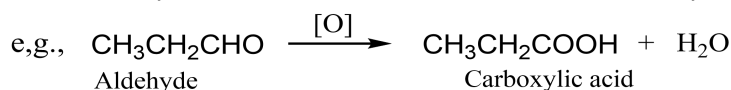
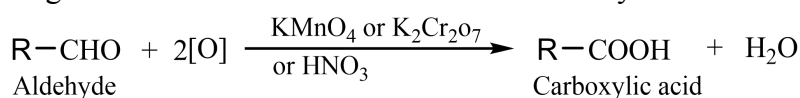
(c) Reduction with HI in presence of red phosphorus:



D. Oxidation reactions: Aldehydes differ from ketones in their oxidation reactions.

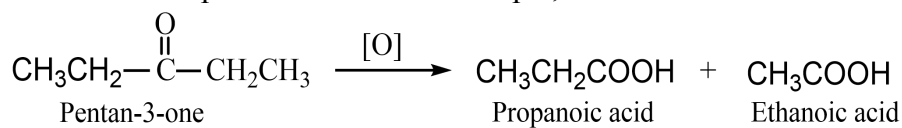
Aldehydes are easily oxidised to carboxylic acids on treatment with common strong oxidising agents like **nitric acid**, KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$, etc. Even mild oxidising agents, mainly **Tollen's reagent** and **Fehling's reagent** also oxidise aldehydes.

1. Oxidation by strong oxidizing agents: On oxidation aldehydes give carboxylic acids containing the same number of 'C' atoms as the aldehyde.

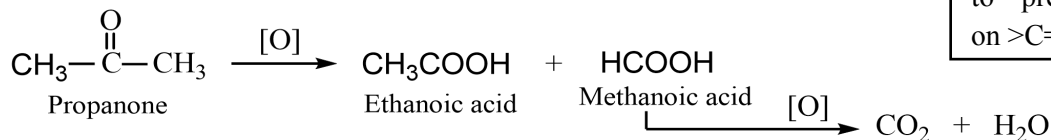


☞ The reason of the easy oxidation of aldehydes is the presence of 'H' atom on the $>\text{C}=\text{O}$ group which can be converted into $-\text{OH}$ group.

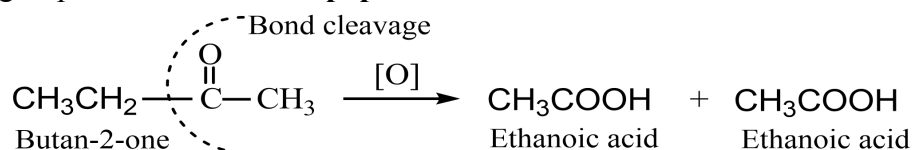
Ketones are generally oxidised under vigorous conditions, i.e., strong oxidising agents (e.g., conc.HNO₃, KMnO₄/H₂SO₄, K₂Cr₂O₇/H₂SO₄ etc.) and at elevated temperatures. Their oxidation involves carbon-carbon bond cleavage to afford a mixture of carboxylic acids having lesser number of carbon atoms than the parent ketone. For example,



➤ -CH₃ group is first oxidized to HCOOH and then further oxidized to CO₂ and H₂O due to presence of 'H' atom on >C=O group in HCOOH.



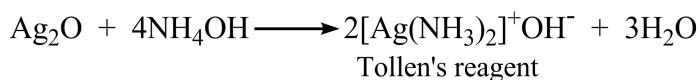
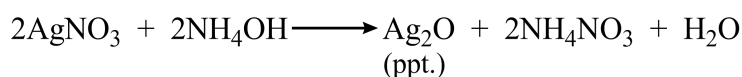
☞ In case of unsymmetrical ketones, the keto group (>CO group) stays preferentially with the smaller alkyl group. This rule is called **popoff's rule**.



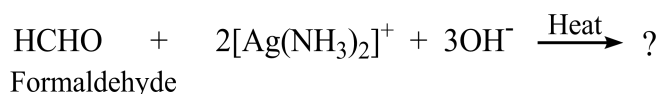
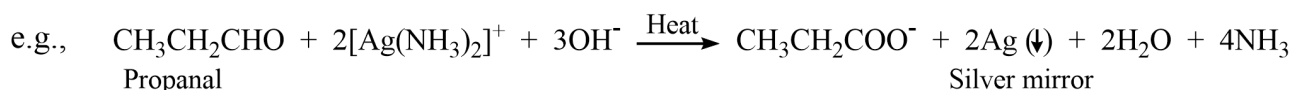
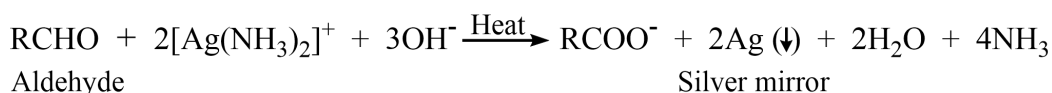
2. Oxidation by mild oxidizing agents (Distinction between Aldehydes and Ketones):

The mild oxidising agents like **Tollen's reagent** and **Fehling's solution** are used to distinguish aldehydes from ketones. Aldehydes are easily oxidized by these reagents but ketones do not.

(i) Tollen's test: Tollen's reagent is freshly prepared ammoniacal solution of **silver nitrate** (AgNO₃).



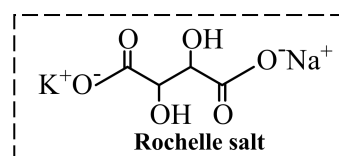
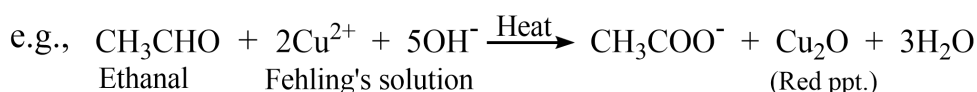
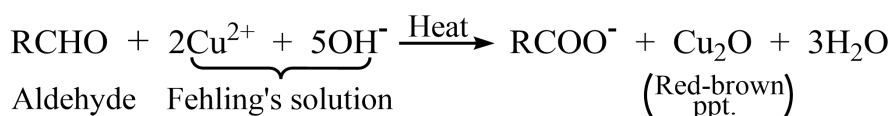
On warming an aldehyde with Tollen's reagent, a bright silver mirror is produced due to the formation of silver metal inside the test tube (**silver mirror**). The aldehydes are oxidised to corresponding carboxylate anion. The reaction occurs in alkaline medium.

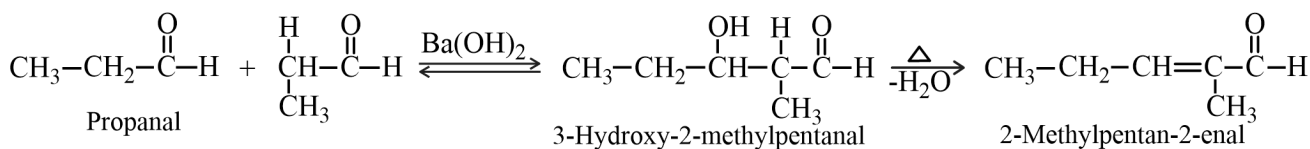


☞ Both **aromatic** and **aliphatic** aldehydes exhibit **Tollen's test** or **Silver mirror test** (i.e., reduce Tollen's reagent).

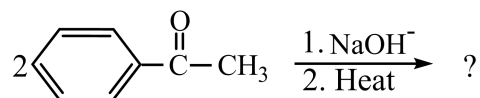
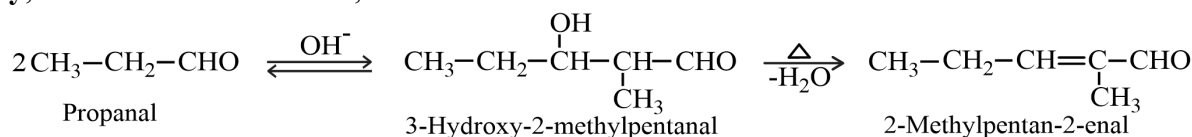
(ii) Fehling's test: Fehling's reagent comprises of two solutions, Fehling solution **A** and Fehling solution **B**. Fehling **solution A** is aqueous copper sulphate and Fehling **solution B** is alkaline sodium potassium tartarate (Rochelle salt). These two solutions are mixed in equal amounts before test.

On heating an aldehyde with Fehling's solution, a reddish brown precipitate is obtained. Aldehydes are oxidised to corresponding carboxylate anion. **Aromatic aldehydes do not respond to this test.**

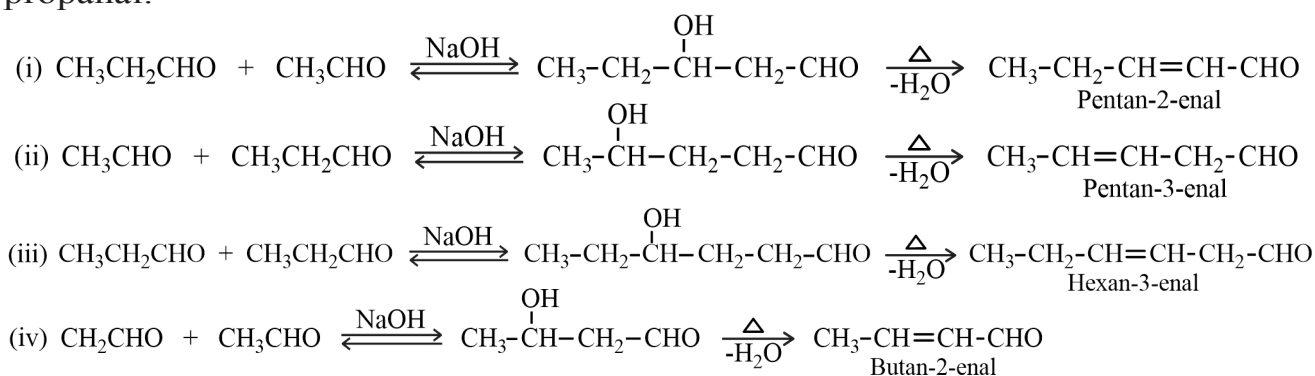




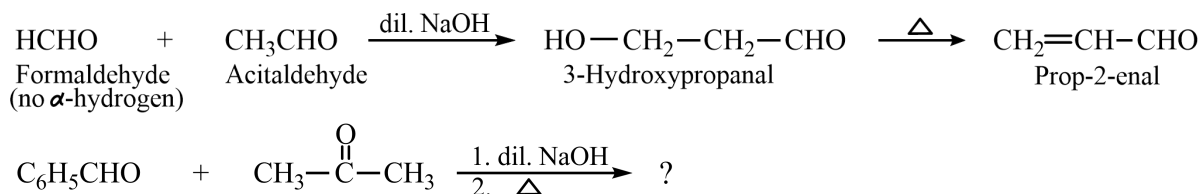
Simply, the reaction is written as,



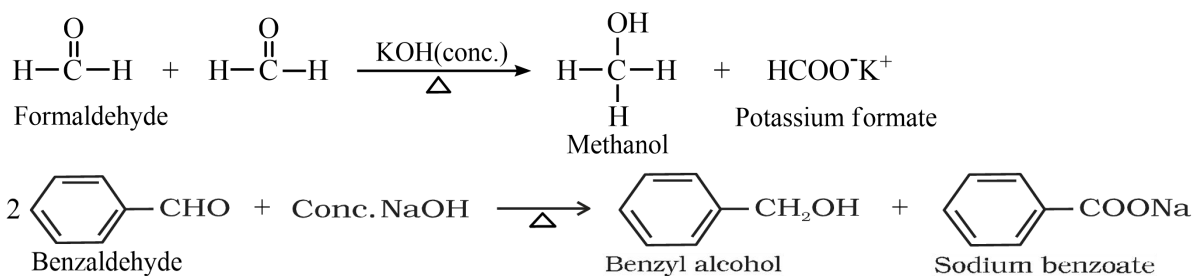
(b) Cross aldol condensation: When aldol condensation is carried out between two different aldehydes and / or ketones, it is called **cross aldol condensation**. If both of them contain α -hydrogen atoms, it gives a mixture of four products. This is illustrated below by aldol reaction of a mixture of ethanal and propanal.



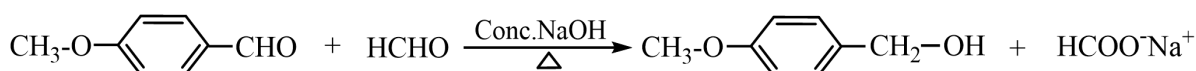
However, if one carbonyl compound does not have α -hydrogen, cross-aldol condensation becomes usefull (i.e., two products are produced). For example,



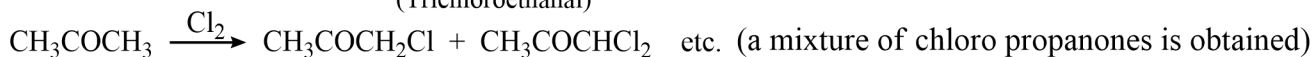
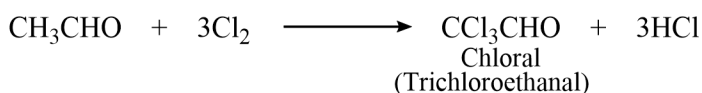
2. Cannizzaro reaction: Aldehydes which do not have an α -hydrogen atom, undergo self **oxidation** and **reduction (disproportionation)** reaction on treatment with concentrated alkali (conc.KOH or NaOH). In this reaction, one molecule of the aldehyde is reduced to **alcohol** while another is oxidized to **salt of carboxylic acid**.



☞ **Cross – Cannizzaro reaction:** Cannizzaro reaction can also take place between two different aldehydes which do not have α – hydrogen atom. For example,

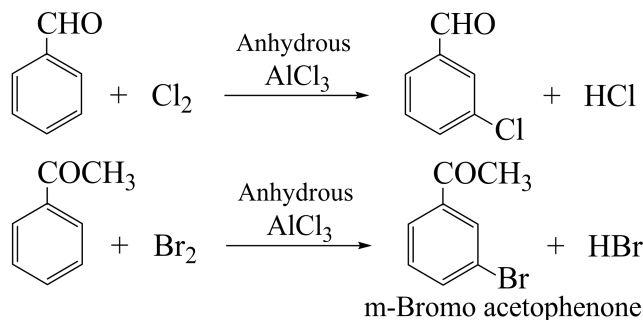


3. Halogenation: Aldehydes and ketones having α -hydrogen undergo halogenations under suitable conditions in which the α -hydrogens are successively replaced by the halogen atoms. For example,

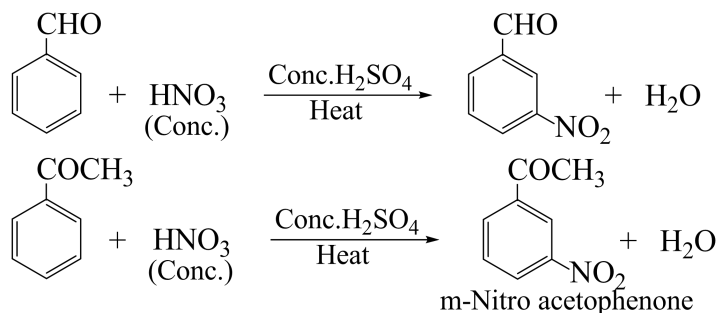


4. Electrophilic substitution reactions of benzene ring:- Aromatic aldehydes and ketones undergo electrophilic substitution at the ring in which the carbonyl group ($-\text{CHO}$ or $>\text{C}=\text{O}$) acts as a **deactivating** and **meta-directing** group. Some examples are,

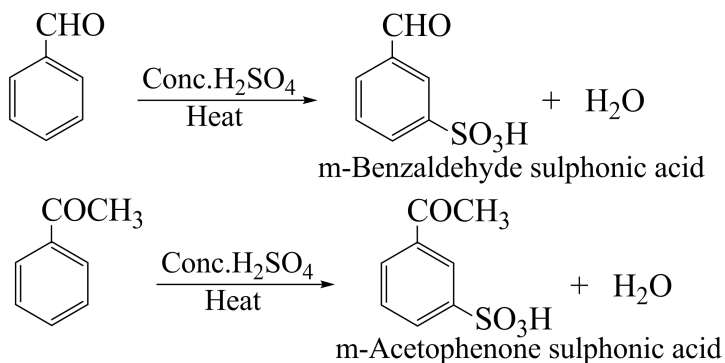
(i) Halogenation:



(ii) Nitration:-



(iii) Sulphonation:-



Question:- Arrange the following compounds in increasing order of their reactivity in nucleophilic addition reactions.

- Ethanal, Propanal, Propanone, Butanone.
- Benzaldehyde, *p*-Tolualdehyde, *p*-Nitrobenzaldehyde, Acetophenone.

Hint: Consider steric effect and electronic effect.

Question: An organic compound (A) with molecular formula $\text{C}_8\text{H}_8\text{O}$ forms an orange-red precipitate with 2,4-DNP reagent and gives yellow precipitate on heating with iodine in the presence of sodium hydroxide. It neither reduces Tollens' or Fehlings' reagent, nor does it decolourise bromine water or Baeyer's reagent. On drastic oxidation with chromic acid, it gives a carboxylic acid (B) having molecular formula $\text{C}_7\text{H}_6\text{O}_2$. Identify the compounds (A) and (B) and explain the reactions involved.

Solution: Compound (A) forms 2,4-DNP derivative. Therefore, it is an aldehyde or a ketone. Since it does not reduce Tollens' or Fehling reagent, (A) must be a ketone. (A) responds to iodoform test. Therefore, it should be a methyl ketone. The molecular formula of (A) indicates high degree of unsaturation, yet it does not decolourise bromine water or Baeyer's reagent. This indicates the presence of unsaturation due to an aromatic ring.

Compound (B), being an oxidation product of a ketone should be a carboxylic acid. The molecular formula of (B) indicates that it should be benzoic acid and compound (A) should, therefore, be a monosubstituted aromatic methyl ketone. The molecular formula of (A) indicates that it should be phenyl methyl ketone (acetophenone). Reactions are as follows:

